ASL Smart Translator

Problem and Motivation

According to the World Federation of the Deaf, there are more than 70 million deaf people worldwide. And yet, according to a study by 2010-2018 National Health Interview it was found that in the USA, sign language use among adults (including those with and without hearing loss) was roughly 2.80%.

Let us zoom into India itself , according to the 2011 Census, the total population of deaf persons in India numbered about 50 lakh. Yet, as reported by the India Sign Language Research and Training Center, the needs of these people were severely ignored. The teaching methods needed urgent attention, while ISL existed, a lot of parents weren't aware of its ability to ease the communication barriers. At that time, there were 300 certified interpreters despite the urgent requirement of it in various fields.

All of this shows us how communication becomes a significant barrier for individuals who rely on sign language on a day to day basis. This is especially felt in fields of healthcare and education. And while alternatives such as text based communication and interpreters exist, they are not practical or sometimes not available for real-time situations.

Recent development in AI and Machine learning is helping in improving communication and making the world a bit more accessible for people with disabilities. An example can be the RayBan Meta Smart glasses that help the blind community, or even the up and coming Google's SignGemma. Similarly, to bridge this gap, we wanted to make a project that can help in recognizing ASL and translating it for individuals with the help of generative AI and LLM models and machine learning models.

Method

The proposed system combines the concepts of machine learning, natural language processing, and computer vision.

The system captures hand gestures using a webcam. It then extracts the hand landmarks using Mediapipe and predicts which gesture it is. This is done using a trained machine learning model.

It then converts the output into a proper sentence and translates it in Hindi.
The overall pipeline is shown in the figure.

The dataset was collected in two ways.

1. First, we used the dataset available on Kaggle called WS_ASL. This contained the entire ASL
2. We recorded our own inputs for certain key phrases, alphabets and numbers. The phrases we recorded our inputs were -

The videos collected were recorded in different lightings, different hand positions and by different people to make the model more fit for real-time situations. This was also done to make sure that the model doesn't learn a bias for a specific person and that there is variation.

Then, we took the CSV values of both datasets. We got the CSV values of the dataset we recorded by using Mediapipe. Mediapipe collected 21 landmarks for each hand and each gesture. These landmarks were mapped in (x,y,z) coordinates.

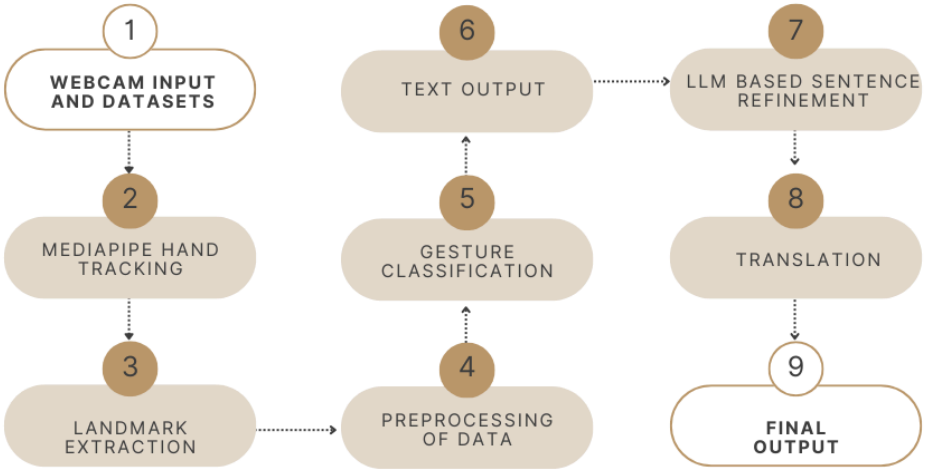
To improve the consistency, normalization was applied in the landmarks. This reduced the effect of hand position and camera distance during the training and inference. It was done relative to a base landmark point on the hand.

These landmarks were then used to train a machine learning classifier which would help us in gesture recognition. We first used Random Forest to train our dataset. While Random forest worked well for phrases and numbers, it did not predict the gestures with a high accuracy for alphabets. To combat this issue, we switched to XGBoost. When XGBoost showed better results overall, we changed our code and stuck with XGBoost.

During real-time inference, the webcam frames are continuously processed using the same landmark extraction and pre-processing pipeline used during data training. The trained model predicts the corresponding ASL alphabet, gesture or number.

After prediction, sentence builder and smoothing is done using LLM's to make sure the gestures form a valid and understandable sentence. It is also translated in Hindi language to make it more accessible for a larger population.

PIPELINE

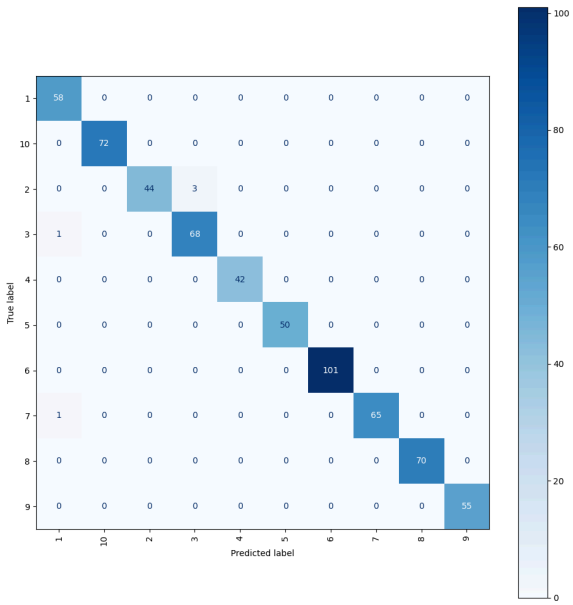

1
WEBCAM INPUT AND DATASETS
2
MEDIAPIPE HAND TRACKING
3
LANDMARK EXTRACTION
4
PREPROCESSING OF DATA
5
GESTURE CLASSIFICATION
6
TEXT OUTPUT
7
LLM BASED SENTENCE REFINEMENT
8
TRANSLATION
9
FINAL OUTPUT

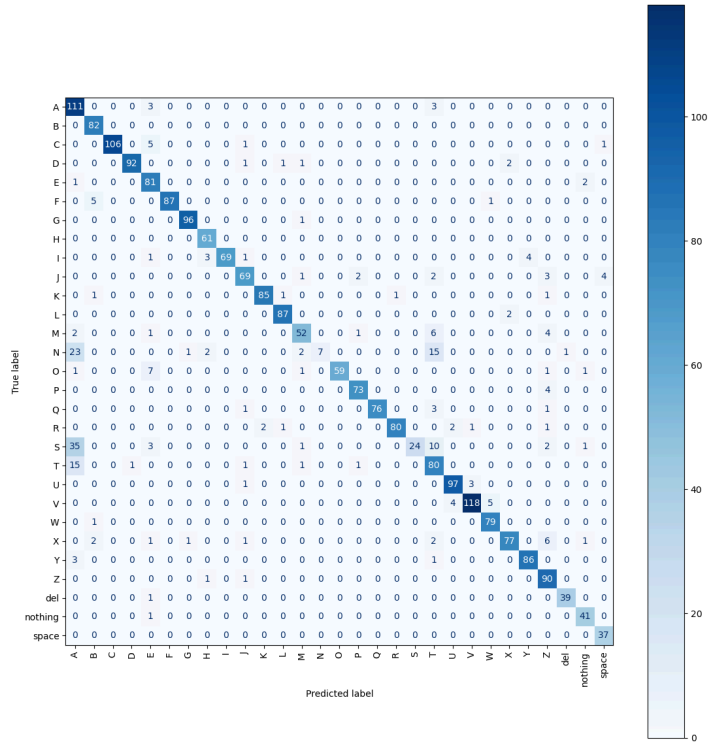
Experiments and Results

A. Experimental Setup
The system was evaluated using landmarks that was generated from the ASL dataset containing alphabets, numbers, and phrases. The dataset was a combination of the CSV values which were recorded by us and evaluated using MediaPipe as well as the CSV values of the dataset found online on Kaggle. The dataset was split into testing and training sets. Two machine learning approaches were tested - Random Forest and XGBoost. Model performance was evaluated using accuracy, precision, recall, F1 score and confusion matrices.

B. Random Forest Results
Initially, Random Forest was used as the primary classifier for gestures. The model showed success for ___, and phrases recognition.
The table below shows the accuracy of the model. While there is high accuracy for numbers, there is lower performance for phrases due to increase in gesture complexity. We have also attached images to show the confusion matrix of alphabets, phrases and numbers.

Dataset	Accuracy	Macro Avg Precision	Macro Avg Recall	Macro Avg F1 score	Weighted Avg F1 score
Numbers	99.21%	0.99	0.99	0.99	0.99
Alphabets	89.66%	0.92	0.89	0.88	0.89
Phrases	74.25%	0.83	0.73	0.75	0.76





C. XGBoost Results

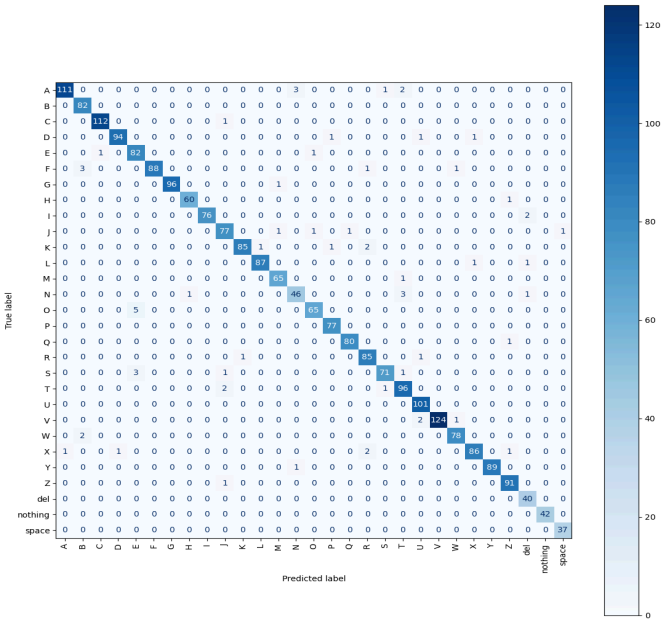
To improve prediction accuracy, XGBoost was later implemented as the primary classifier. Compared to Random Forest, XGBoost produced significantly more stable and accurate predictions across all datasets.

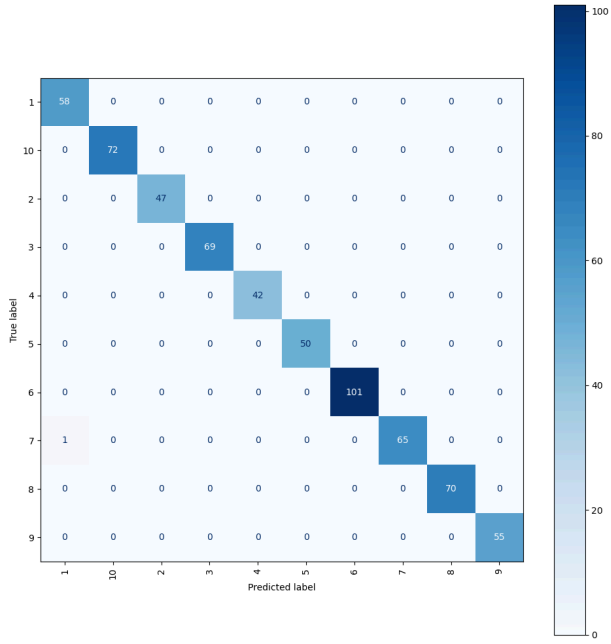
The XGBoost model achieved substantial improvements in both alphabet and phrase recognition tasks. The confusion matrices showed significantly fewer misclassifications compared to the Random Forest model. The use of x, y, and z landmark coordinates also improved recognition by capturing depth-based spatial information.

We can see the accuracy and the confusion matrix of the number and alphabet dataset attached below

Dataset	Accuracy	Macro Avg Precision	Macro Avg Recall	Macro Avg F1 Score	Weighted Avg F1 Score
Alphabets	97.28%	0.97	0.97	0.97	0.97
Numbers	99.84%	1.00	1.00	1.00	1.00

Phrases	95.84%	0.96	0.96	0.96	0.96





D. Model Comparison and Ablation

A comparison between Random Forest and XGBoost was performed to evaluate the effect of model selection on recognition performance. As seen by the table below, the comparison shows that XGBoost significantly outperformed Random Forest across all major tasks. XGBoost handled complex landmark relationships and subtle finger positioning differences more effectively, resulting in improved prediction stability and higher overall accuracy.

Task	Random Forest	XGBoost
Alphabet Recognition	89.66%	97.28%
Landmark Recognition	74.25%	95.84%

Analysis

One major strength of the project was using Mediapipe for the real-time hand landmarks instead of using raw image frames. This allowed the spatial hand information to be preserved while reducing the computational complexity.

The system performed reasonably well for number and phrase recognition during real time testing. The text output and sentence refinement pipeline also made it easy to understand context even if complete sentences weren't being signed.

During development, we used multiple approaches for gesture recognition. We initially used a CNN based image classification, but that wasn't giving us accurate predictions because it wasn't tuned to the real-time settings. The settings being the variation in light, the hand orientations etc.

We then shifted to a landmark based classification approach. We did so by applying Random Forest. This had improved the prediction significantly, especially for phrases and for numbers. However, our accuracy for alphabets was not satisfactory.

To further improve the performance, we then implemented XGBoost as the primary classifier. Compared to Random Forest, this handled data more effectively and hence gave better predictions during testing.

While there is a lot of progress, the system still has more improvements to make. Currently, the model is able to translate and understand our ASL signs effectively. However, considering the fact that none of the members of the team are fluent in ASL, and hence the speed of signing will vary for someone who is, we are yet to see if it would be able to capture dynamic and continuous sentence-level ASL translation with the same accuracy.
Further, the real-time prediction can still be affected by the lighting conditions and camera quality.

Overall, the project shows how combining computer vision, machine learning, LLM based text processing can help reduce the communication barriers for sign language users in certain real time situations. While we have a more focused dataset currently, it provides a practical foundation for more advanced sign language translation systems in the future.

Contributions

1. Shrivatsh K - Collecting Dataset, Pre-processing the data, Training and Testing the Model using the
2. Chirag Goyal - Collecting Dataset for alphabet, numbers and dataset using Mediapipe as well as converting into CSV.
3. Meesha Maheshwari - Using LLM to refine the sentence and translate the text, using API key to access LLM, making the UI/UX
4. Vinay - Making the presentation, helped in collecting phrases for dataset
5. Manohar - Making the presentation, helped in collecting phrases for dataset

Bibliography

1. https://islrtc.nic.in/about-us/
2. https://pubmed.ncbi.nlm.nih.gov/36423340/